

INTELLIGENT BREAST CANCER IDENTIFICATION SYSTEM USING DEEP TRANSFER LEARNERS WITH SVM CLASSIFIER MODELS

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Abstract:

Breast cancer (BC) is the most common disease in the medical field. It is the second leading cause of cancer-related deaths after lung cancer, especially in women. Early detection and treatment of the disease are essential for the survival of patients. Ultrasound image analysis is one of the procedures for breast cancer identification. We collected ultrasound cancer image data from the Kaggle Data Repository. Nowadays, automated intelligent cancer detection through image datasets using deep learning methods is one of the most efficient techniques. This research analyses and extracts the features of breast ultrasound cancer images using deep transfer learning neural networks such as Inception V3, VGG-16, and VGG-19 networks. The experimental transfer learners produce the image features (Inception-v3, VGG-16 net, and VGG-19 net). After that, we use the 10-cross folders SVM classifier with different kernels, such as the radial bias (RBF) function, polynomial, sigmoid, and linear, to classify into three categories that are benign, malignant, and normal. According to our observations, the Inception-v3 transfer learner produces quality features for classification. Its classification accuracy is higher than that of other transfer learners classified by the SVM with all kernels. Interestingly, the SVM polynomial kernel performs better for all features of transfer learning than the remaining SVM kernels. According to our findings, the Inception-v3+SVM (polynomial) performance is very high, with a 0.944 AUC and 87.44% classification accuracy, higher than other experimental models.

Keywords: Breast Cancer, Deep Learning, Feature Extraction, Inception-v3, SVM, Transfer Learning

1. Introduction

Breast cancer (BC) represents the most suspicious disease in the world, defined by the growth of abnormal cells uncontrollably in the breast. The BC is a kind of tumour that arises from the breast muscle. It most commonly happens within the lobules or milk ducts that source the milk to the epithelial duct. [1] The American Cancer (ACS) Society estimates that approximately 2.3 million new cases were reported worldwide in 2020; that is 11.7% of all new cancer diagnoses. The ACS found that 13% of women will produce breast cancer during their lifetime. This disease increased year by year (2008 to 2017 analysis) by nearly 0.5 percent. After lung cancer, the breast cancer is the most common cancer that causes death in women. Among women between the ages of 35 and 54, it is also the leading cause of cancer death. According to WHO (World Health Organization) reports, 685000 people globally died in 2020 impact to breast cancer, which affected millions of women (nearly 2.3). The most common cancer in the world as of 2020 end of the year was breast cancer, which had been diagnosed in women (7.8 million) in the previous five years [2-3]. As per statistical analysis, the analyses suggested some of the points to women. If they choose, women between the ages of 40 and 44 should be able to start getting annual mammograms, which are x-rays of the breast. Every year, women between the ages of 45 and 54 should get mammograms. Women over the age of 55 should get mammograms every two years, or they can keep getting screened every year. As long as a woman is in good health and is expected to live at least another ten years, screening should continue. All women ought to be aware of the known advantages, drawbacks, and dangers of breast cancer screening.

An ultrasound creates an image of the tissues deep within your breast using sound waves [4]. Your doctor can use an ultrasound to tell the difference between a solid mass, such as a tumour, and a benign cyst. A mammogram, an imaging test, is the most common way to see below the surface of your breast [5]. Annual mammograms are performed on women over 40 to check for breast cancer. A breast biopsy or an MRI are the two other suggested tests [6]. The table shows the Breast cancer stages analysis in detail. It describes diagnosis systems, prevention and treatment, affected breast area, and so on [33].

Table 1: Stages of Breast Cancer, Diagnosis, Symptoms, Protection and Treatment

Stages	Affected area of Breast	Diagnosis	Precautions and Treatment
Stage 0	The disease doesn't hurt. This indicates that it has not emerged from our breast ducts.	Regular Self-testing. As per Symptoms, Family history and Age consult the doctor.	Avoid using any tobacco materials. Maintain a healthy weight Engage in regular exercise. Fruits and vegetables for a healthy diet [7].
Stage 1	The cells of cancer have reached the breast tissue nearby.	Regular Self-testing. Make an appointment with your doctor if you notice spot or an unusual	Chemotherapy, hormone therapy, radiation, or targeted therapy. Be aware of ourselves, your

		lump in your breast or other signs of breast cancer. Other imaging tests.	family's past, and your dangers. Get cancer screening tests and check-ups.
Stage 2	At this stage, tumours can be anywhere from 2 to 5 centimetres across and may or may not affect the lymph nodes in the vicinity.	Symptoms, Imaging tests like Ultrasound, Mammogram, Magnetic resonance (MRI) imaging	Surgery is the most commonly utilized therapy for breast cancer. Many individuals get supplemental therapies like as chemotherapy, hormone therapy, radiation, or targeted therapy [8].
Stage 3	The cancer has spread beyond its original location at this point. Typically, breast cancer at stage III is referred to as locally advance [9].	Symptoms, Mammogram, Magnetic resonance (MRI) imaging, and Breast biopsy. Some other essential tests prescribed by doctors.	Surgery, Chemotherapy, hormone therapy, radiation, or targeted therapy. Avoid using any tobacco materials. Maintain a healthy weight. Engage in regular exercise. Fruits and vegetables for a healthy diet.
Stage 4	Breast cancer tumours can be of any size. Cancer cells from it have migrated to local and regional lymph nodes, in addition to lobules [10].	Symptoms, Mammogram, Magnetic resonance (MRI) imaging, Ultrasound, and Breast biopsy. Some other essential tests prescribed by doctors.	Surgery, Chemotherapy, hormone therapy, radiation, or targeted therapy. Avoid using any tobacco materials. Maintain a healthy weight. Engage in regular exercise. Fruits and vegetables for a healthy diet. Avoid drinking alcohol at all costs

One of the emerging fields in medical applications is image processing. Medical imaging is helpful for diagnosing and detecting diseases early on. Pre-processing, optimization and Deep Learning methodologies are used to detect breast cancer at an earlier stage. This research aims to develop an ultrasound image of a breast cancer identification system using deep transfer learners with SVM classifier models[31]. The system will use deep learning algorithms to extract features from the images and then use a Support Vector Machine (SVM) classifier to classify the images into benign, malignant or normal. The system will be trained on a dataset 10-folder cross validation of breast ultrasound images and then tested on a separate dataset. The performance of the models was evaluated using accuracy, precision, recall, and F1 score metrics. The system will be able to accurately identify breast cancer in ultrasound images and provide a reliable diagnosis[30][32].

The remaining sections of the paper is presented as

- Section 2: literature review of various research works related to breast cancer on ML and imaging classification with DL concepts
- Section 3: proposal model and materials represent the working process of the images of breast cancer identification through novel approaches and models.
- Section 4: results analysis depicts the analysis of the pre-trained transfer learners for feature extraction and apply the various kernels of SVM models performances, and simulation results.
- Section 5: analysis of comparison presents the detailed discussions and compares the proposed model against other works related to this research work.
- Section 6: conclusion reports and describes the limitations of the research and future works related to breast cancer.

2. Literature Survey

In this section, we promote different research works on breast cancer, particularly those analyzed using breast cancer images. To do this, we reviewed various papers from reputed journals and presented some of the research works. The use of Computer aided (CAD) detection systems frequency enhanced in the recent year for diagnosis of medical image analysis. According to doctors' psychoanalysis, the breast anatomy, particularly dense ones, as well as the probability of breast cancer and tumor growth, vary significantly depending on race. Alsolami et al. (2021) [11] described the mammogram image dataset of breast cancer. In this paper, they analyzed the KAU-BCMD data set in detail that re-examined 1416 cases of images 5662 (both left and right breasts). Apart from 205 patients' 405 images of ultrasound are analyzed. There were three different radiologists annotated the data set[26].

Zhu et al. (2020) [12] described elaborately deep learning versus health care. Mainly, they focussed on the images of cancerous diagnosis and predictions. In this review, they analyzed the accuracy of prognosis prediction cancers with deep learning methodologies, cost reduction, and many issues of applications in cancer diagnosis systems[27]. They suggested the best DL methodologies for the identification and prediction of cancers for the survival of the patients. Tufail et al. (2021) [13] reviewed many research papers on cancer predictions, diagnosis challenges, and trends in deep learning. They described more image data sets related to breast cancers with deep learning methodologies. As per their findings, the argumentation of the data is significant to the prediction and diagnosis of cancerous. Kourou et al. (2021) [14] reviewed the last five years of research works considering the databases DBLP and PubMed[28]. As per comparisons, the analysis was in reinforcement learning, deep learning, and AI/ML applications related to cancerous. Tang et al. (2019) [15] reviewed and elaborated computational biology and bioinformatics for DL methodologies. In this review, they studied frequently used DL model similarities, basic structures, and their application. In this scenario, they describe the disadvantages and advantages of suicide and failure models in detail[29].

Table 2: Stages of Breast Cancer, Diagnosis, Symptoms, Protection and Treatment

Rf. No .	Ref. of Author (year)	Description	Data set and Methodologies	Highlights
[16]	Lee et al. (2010)	Analysed breast imaging for cancer screening use of ultrasound, mammography, MRI and other detection techniques	Imaging analysing and screening Models	Screening breast cancer with all imaging methodologies
[17]	Wu et al. (2013)	Analysis and classification of ultrasound breast images using SVM and Genetic Algorithms	Use total 210 images of Breast (120 benign 90 malignant images) Models are PCA for feature selection SVM and SVM+GA for classification	The proposal model SVM+GA performs well than other experimental models.
[18]	Housein et al. (2021)	Examined the most recent research on the detection and classification of breast cancer, which included 760 studies that made use of a variety of machine learning and deep learning strategies across five image modalities.	Used ML models SVM, DT, KNN, NB Network, and ANN and Deep Learning models like Transfer learner on Mammograms, Ultrasound, and MRI Datasets.	Studied 760 breast cancer research papers and conclude DL models are very efficient for image classification.
[19]	Rezaei, (2021)	Reviewed different breast cancer image datasets and analyzed different methodologies on images like segmentation, classification and detection	Use all breast cancer imaging datasets Mammograms, Ultrasound, MRI Datasets for analysis and ML and DL model for classification	Public datasets and the diagnosis of breast cancer are also introduced. The main gaps and problems in the current methods are discussed.

[20]	MurtiRawat et al. (2020)	Classification using ML models for breast cancer diagnosis.	ML algorithms such as KNN, Logistic Regression, Ensemble learning	Proposal model Ensemble learning model is gives high accuracy than other models
[21]	Vakanski et al. (2020)	Image Segmentation analysis on Breast Ultrasound images	510 Breast Ultrasound images U-Net architecture and learns feature representations	U-Net model achieved good accuracy.
[22]	Luo et al. (2022)	tumor region segmentation and classification based analysis on Breast Ultrasound images	Deep Learning methodologies.	DCNN is best classifier for Breast cancer dataset
[23]	Fei et al. (2021)	Novel Deep Transfer learning models applied on imbalanced Breast Ultrasound images	Apply Deep Transfer learner novel models like doubly supervised parameter (DSPTC) transfer classifier.	As per comparison the DSPTC model presents the good accuracy for imbalance image dataset.

3. Materials and Models

In this, we describe the proposed model and methodologies for breast cancer identification. The transfer learning models like Inception V3, VGG-16, and VGG-19 have using for feature extraction. The classifier SVM with various kernels (RBF, Linear, Polynomial, and sigmoid) used the classification of breast cancer classes (Malignant, Benign, and Normal).

Proposal Model

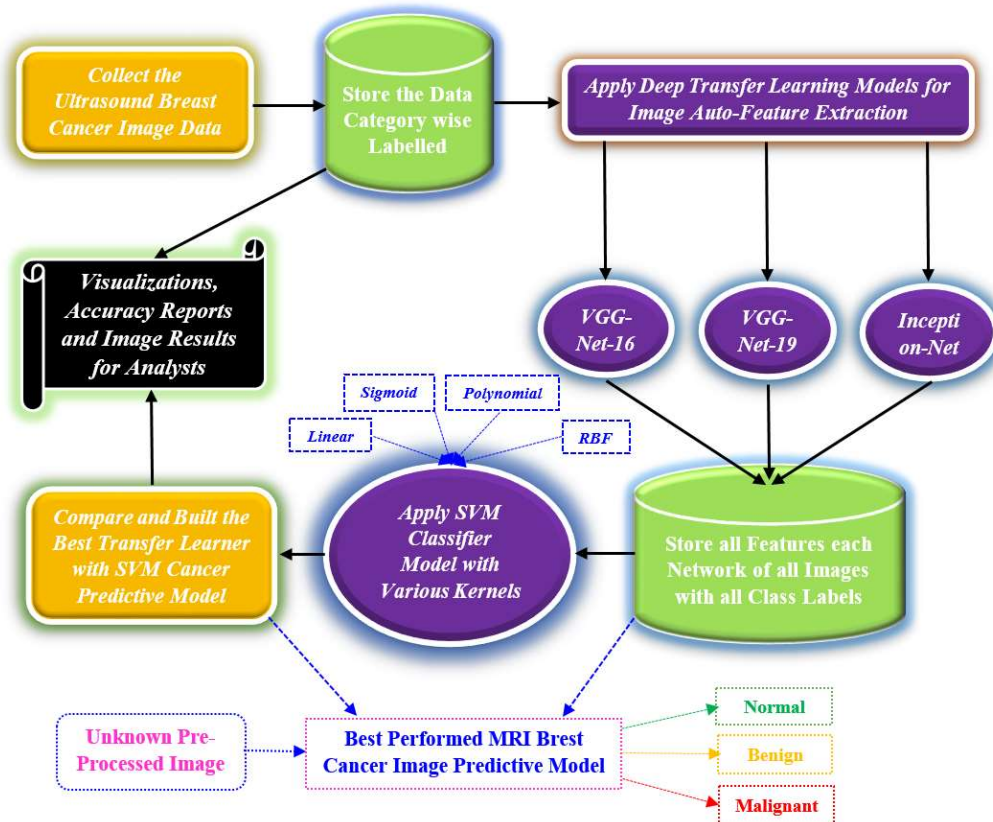


Figure 1: Proposal model for Breast cancer identification using Transfer Learners + SVM classifier

Figure 1 shows the proposed model that describes the overall technical process for detecting breast cancer using image data set. Firstly, collect the Ultrasound Breast cancer Image data and store it in three category-wise folders: Benign, Malignant, and Normal. Extract the features using deep transfer learning models VGG-16, VGG-19 and Inception v3, and store their feature values individually for classification with an SVM classifier. Split the data with 10-folding method and then apply the SVM classifier with various kernels (Sigmoid, Linear, RBF, and Polynomial). Compare the each model performance and choose the best performed model for the predict the breast cancer. After declaring the best performing model, predict the unknown ultrasound breast cancer images. All the results will be sent to the analysts for analysing the results.

Data Set Description:

The Breast Cancer Ultra sound imaging data is collected from Kagle data repository. The data is total 780 images in 3 categories that are Benign 440, Malignant 210 and Normal 130. The figure 2 shows the different categories of Breast cancer. The figure 2(A) describes the benign category breast cancer, figure 2(B) shows the Malignant and Normal Breast ultra sound Images (shown in 2(C)).

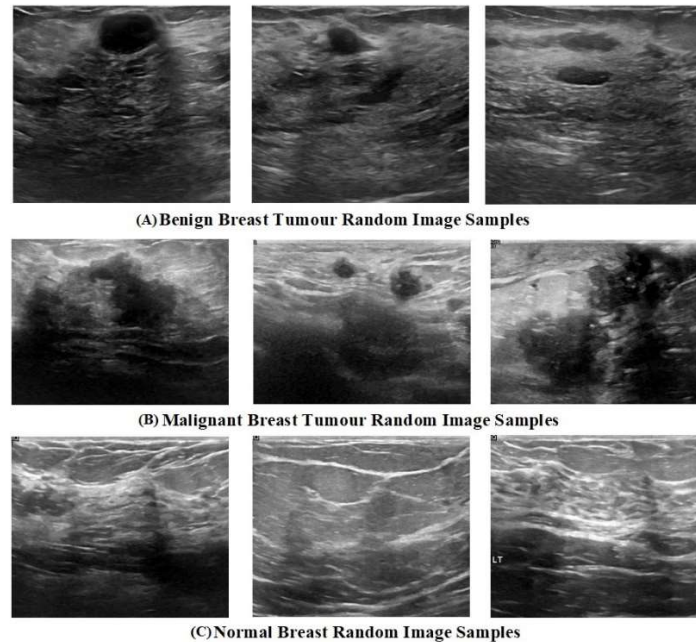


Figure 2: Categories of Breast cancer Images

Transfer Learning (TL)

Transfer learning (TL) is a technique for reusing a previously trained model that is used just for another task. Pre-training is the initial training step. Pre-training "teaches" the conceptual framework's general overall features, while end of the training stage "notifies" features particular to our data. As per experimental truths that normal CNNs take the long period of time for processing the large image datasets. Instated of this the TL Networks are very useful for time consuming process. Another important thing that it is fast and accurate than other custom models. Figure 3 shows the Transfer learning working process pre-trained network and applied saved network for the cancer or other images.

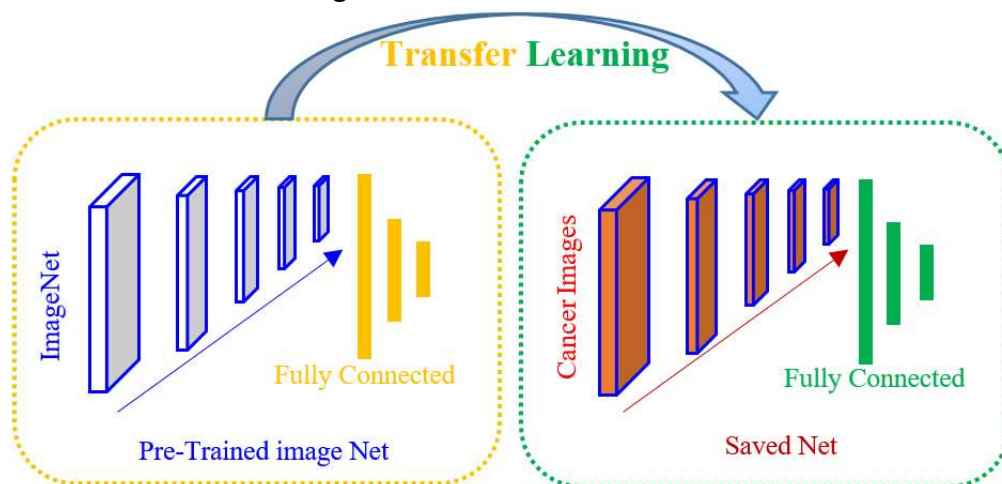


Figure 3: Transfer learning working process

VGG network constructions and structure with deeper networks and smaller filter sizes like 3 by 3 conv. VGG-16 architecture is variant of VGG-Net that it contains layers sixteen and built with

convolution, pooling and fully (FC) connected layers. Primarily, it is the simplest convolutional filter size of it (3 by 3 conv.) that scans a small portion of the surrounding pixels. As a direct consequence, the network has discovered detailed feature representations for just a differing set of images. The network accepts 224-by-224 size image input. The detailed expiation of VGG-16 is shown in figure 4(A).

The figure 4(B) represents the VGG-19 architecture. VGG-19 is a deep CNN consisting of 19 layers, including convolutional layers, pooling layers, and fully connected layers. Here's a brief overview of the VGG-19 architecture: the Input Layer accepts input images with a fixed size of 224x224 pixels. The Convolutional Layers, which are the first layers in the network, consisting of 2D convolutional filters that extract features from the input image. There are a total of 16 convolutional layers in VGG-19. The Pooling Layers, which follow the convolutional layers, are five max pooling layers that down-sample the feature maps and help reduce the dimensionality of the data.

The Fully Connected Layers, which are the final layers of the network, are fully connected layers that perform classification on the extracted features. There are three fully connected layers, with the last layer outputting a probability distribution over the classes in the ImageNet dataset. Suppose we have an image that we want to classify using VGG-19. We would first resize the image to 224x224 pixels to match the input size of the network. Then, we would pass the image through the network, and the convolutional layers would extract various features such as edges, corners, and textures. The max pooling layers would down-sample the feature maps, and the fully connected layers would perform classification on the extracted features. Finally, the output of the last layer would give us a probability distribution over the 1000 classes in the ImageNet dataset, which includes our image as one of the classes. We would select the class with the highest probability as the predicted class for the input image.

Image Input	224x224x3 images with 'zero-centre' normalization
Convolution	64 3x3x3 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	64 3x3x64 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Max Pooling	2x2 max pooling with stride [2 2] and padding [0 0 0 0]
Convolution	128 3x3x64 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	128 3x3x128 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Max Pooling	2x2 max pooling with stride [2 2] and padding [0 0 0 0]
Convolution	256 3x3x128 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	256 3x3x256 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	256 3x3x256 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Max Pooling	2x2 max pooling with stride [2 2] and padding [0 0 0 0]
Convolution	512 3x3x256 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	512 3x3x512 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	512 3x3x512 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Max Pooling	2x2 max pooling with stride [2 2] and padding [0 0 0 0]
Convolution	512 3x3x512 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	512 3x3x512 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Max Pooling	2x2 max pooling with stride [2 2] and padding [0 0 0 0]
Fully Connected	Connected 4096 fully connected layer
ReLU	ReLU
Dropout	50% dropout
Fully Connected	Connected 4096 fully connected layer
ReLU	ReLU
Dropout	50% dropout
Fully Connected	Connected 4096 fully connected layer
Softmax	softmax
output	Classification Output cross-entropies with 'tench' and 999 other classes

(A) VGG-16 Net

Image	Input 224x224x3 images with 'zero centre' normalization
Convolution	64 3x3x3 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	64 3x3x64 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Max	Pooling 2x2 max
Convolution	128 3x3x64 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	128 3x3x128 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Max	Pooling 2x2 max
Convolution	256 3x3x128 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	256 3x3x256 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	256 3x3x256 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	256 3x3x256 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Max	Pooling 2x2 max
Convolution	512 3x3x256 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	512 3x3x512 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	512 3x3x512 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	512 3x3x512 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Max	Pooling 2x2 max
Convolution	512 3x3x512 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	512 3x3x512 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Convolution	512 3x3x512 convolutions with stride [1 1] and padding [1 1 1 1]
ReLU	ReLU
Max	Pooling 2x2 max
Fully	Connected 4096 fully connected layer
ReLU	ReLU
Dropout	50% dropout
Fully	Connected 4096 fully connected layer
ReLU	ReLU
Dropout	50% dropout
Fully	Connected 1000 fully connected layer
Softmax	softmax
Out-Put	1000 features

(B) VGG-19 Net

Figure 4: Transfer Learners VGG-16 and VGG-19 Structures

The figure 5 Inception V3 is a convolutional neural network architecture used for image classification. It consists of several layers including convolutional layers, pooling layers, and fully connected layers. The architecture also employs "inception modules," which are composed of parallel convolutional layers with different filter sizes to capture features at different scales. The input to the Inception V3 network is a 299x299x3 RGB image. The first layer of the network is a convolutional layer with 32 filters, followed by a second convolutional layer with 32 filters. These two layers are followed by a max pooling layer, which reduces the spatial dimensions of the input. Next, the network employs several inception modules that are designed to capture different scales of features. Each inception module consists of parallel convolutional layers of different sizes, with pooling and concatenation operations that allow the network to learn a variety of feature maps. After several inception modules, the network ends with global average pooling, followed by a fully connected layer and a softmax output layer for classification. Additionally, the Inception V3 architecture includes auxiliary classifiers, which are used during training to help the network learn features at different depths in the network.

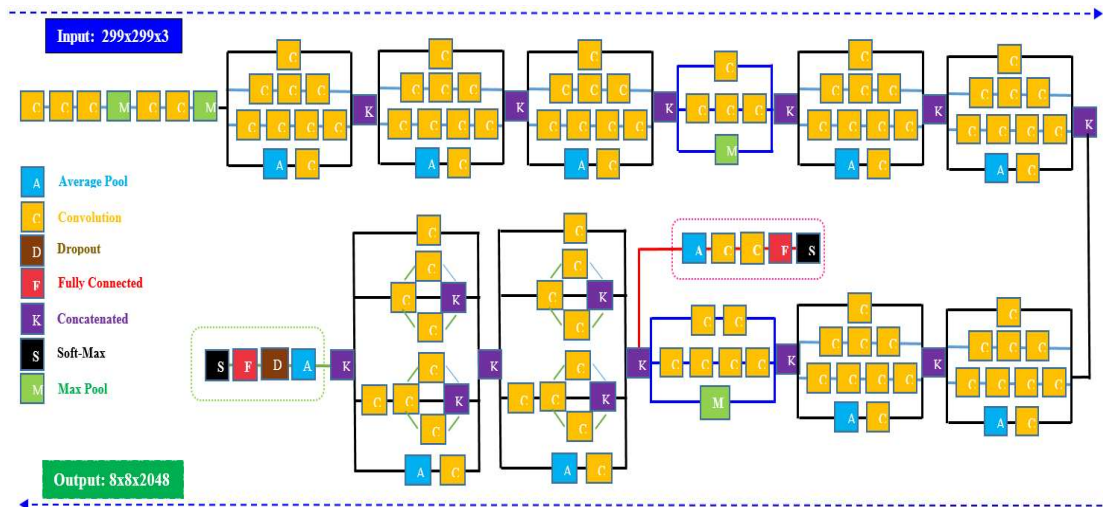


Figure 5: Detailed model of Inception V3 Transfer Learner

SVM Classifier

SVM (Support Vector Machine) is a machine learning model employed in classification and regression analysis. It works by finding the best hyperplane that separates the data into different classes. The margin is determined by the distance between the hyperplane and the closest data points from each category. The SVM algorithm aims to find the hyperplane that maximizes this margin, which leads to better generalization and lowers overfitting. The SVM model can use a kernel function to map the input data to a higher-dimensional space and find a hyperplane. There are several kernel functions, including linear, polynomial, Gaussian (also known as the radial basis function or RBF), and sigmoid. The choice of kernel function depends on the data and the problem at hand. Training the model: The SVM model training involves finding the hyperplane that separates the data into two classes while maximizing the margin. The optimization problem minimizes the classification error subject to a constraint on the margin. Tuning hyperparameters: the choice of hyperparameters can affect the performance of the SVM model, such as the kernel type, regularization parameter, and kernel bandwidth. We can tune the hyperparameters using cross-validation on a validation set to find the best combination that minimizes the test error. Once they have trained and tuned up the SVM model, they can test it by mapping new data points to the same higher-dimensional space and determining which side of the hyperplane they fall on. The SVM model algorithm provides a powerful and flexible solution for a wide range of classification and regression problems. The algorithm's ability to handle non-linearly separable data and its high accuracy make it a popular choice for many real-world applications.

4. Results and Discussions

4.1. Inception V3 (IV3) + SVM classification analysis:

SVM (RBF)		Predicted			Total
Class		Benign	Malignant	Normal	
Actual	Benign	396	39	5	440
	Malignant	41	161	8	210
	Normal	20	5	105	130
Total		457	205	118	780

(A) Inception V3+SVM(RBF)

SVM (Linear)		Predicted			Total
Class		Benign	Malignant	Normal	
Actual	Benign	396	36	8	440
	Malignant	54	148	8	210
	Normal	18	5	107	130
Total		468	189	123	780

(B) Inception V3+SVM(Linear)

SVM (Sigmoid)		Predicted			Total
Class		Benign	Malignant	Normal	
Actual	Benign	364	68	8	440
	Malignant	49	154	7	210
	Normal	25	12	93	130
Total		438	234	108	780

(C) Inception V3+SVM(Sigmoid)

SVM (Polynomial)		Predicted			Total
Class		Benign	Malignant	Normal	
Actual	Benign	407	29	4	440
	Malignant	43	160	7	210
	Normal	12	3	115	130
Total		462	192	126	780

(D) Inception V3+SVM(Polynomial)

Figure 6: Inception V3 + SVM Confusion matrix Analysis

Figure 6 shows the confusion matrices of the InceptioV3+SVM (kernel) 10-fold cross validation models. In this analysis, the extracted features from Inception V3 transfer learning were classified using an SVM classifier with various Kernels. The IV3 + SVM with polynomial kernel performs well for all classes (Benign, Malignant, and Normal) shown in Figure 6 (D). The Normal category images are classified as 115 out of 130 images with near accuracy of 97%, and 160 malignant images are classified correctly out of 210 (nearly 89% accuracy). Figure 6 (A) presents the IV3+SVM with kernel RBF and the class name Normal correctly near 95%. Figure 6 (B) shows the IV3+SVM with kernel Linear classifies the class name Normal, which correctly classified 107 out of 130, 396 benign class images out of 440, and 148 images out of 210. Figure 6 (C) shows the IV3+SVM with kernel sigmoid classifies the class 'normal' correctly 93 out of 130, 364 benign class images out of 440, and 154 images out of 210.

The Table 3 shows the performance parameters like classification accuracy (CA), precision, F-1 values, Recall, and AUC of IV3+SVM (RBF, Polynomial, linear and sigmoid kernels). It describes the performance values of breast cancer classes normal, malignant, and benign individually. As per analysis, some intrusting findings are observed that are 1. IV3 transfer learning developing features are very accurate in the cancer class name "normal" (SVM all kernel models are performed above 90% of accuracy for the class name "normal"); 2. In particular, SVM with kernel polynomial classifies all classes more highly than other kernels+SVM models. In this, the SVM+sigmoid performs very low than other models.

Table 3: Inception v3+SVM (Kernel) Performance Parameters Analysis on target class

Inception v3+SVM(Kernel)	Target Class	CA	F1	Precision	Recall	AUC
SVM (RBF)	Benign	0.8654	0.8829	0.8665	0.9000	0.9366
	Malignant	0.8808	0.7759	0.7854	0.7667	0.9187
	Normal	0.9513	0.8468	0.8898	0.8077	0.9724
	All (Avg.)	0.8487	0.8481	0.8486	0.8487	0.9359

SVM (Polynomial)	Benign	0.8872	0.9024	0.8810	0.9250	0.9400
	Malignant	0.8949	0.7961	0.8333	0.7619	0.9281
	Normal	0.9667	0.8984	0.9127	0.8846	0.9775
	All (Avg.)	0.8744	0.8731	0.8734	0.8744	0.9440
SVM (linear)	Benign	0.8513	0.8722	0.8462	0.9000	0.9199
	Malignant	0.8679	0.7418	0.7831	0.7047	0.9094
	Normal	0.9500	0.8458	0.8699	0.8231	0.9621
	All (Avg.)	0.8346	0.8327	0.8331	0.8346	0.9252
SVM (Sigmoid)	Benign	0.8077	0.8292	0.8311	0.8272	0.8077
	Malignant	0.8256	0.6937	0.6581	0.7333	0.8256
	Normal	0.9333	0.7815	0.8611	0.7154	0.9333
	All (Avg.)	0.7833	0.7847	0.7895	0.7833	0.7833

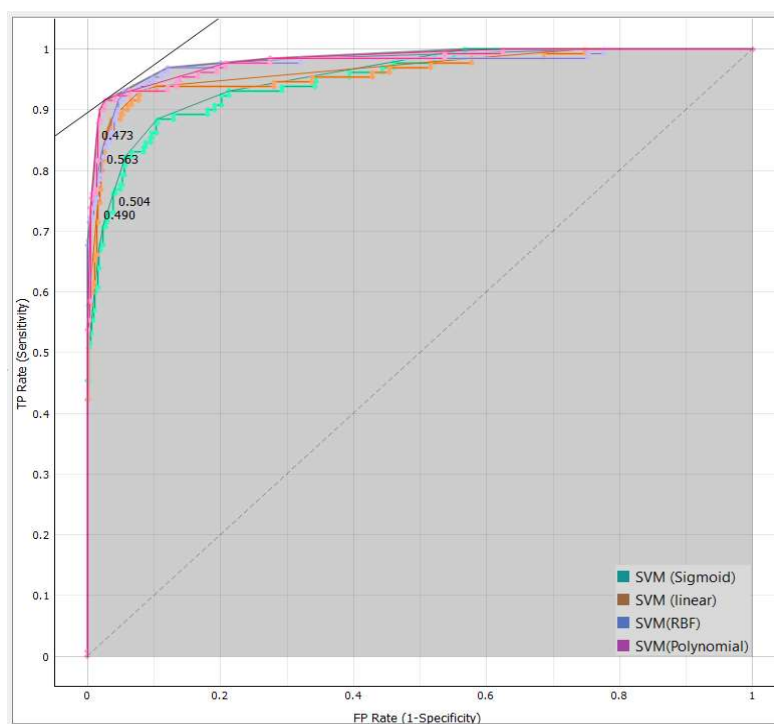


Figure 7: Inception V3 + SVM (Kernels) AUC under ROC Analysis

The figure 7 shows AUC (Area under the ROC Curve) of the SVM (RBF) represents the blue colour, SVM (Polynomial) indicated by violet, SVM (linear) chooses the brown and SVM (Sigmoid) in green, and the AUC values are 0.9359, 0.9440, 0.9252 and 0.7833 respectively. As per AUC ROC analysis, all kernel SVM models performs well except sigmoid SVM mode. The SVM (polynomial) performs highly than other SVM moels with 0.944.

4.2.VGG-19 +SVM classification analysis:

Figure 8 shows the confusion matrices of the VGG-19+SVM (kernels) 10-fold cross validation models. As per the analysis, the extracted features from VGG-19 transfer learning were classified using an SVM classifier with various kernels like RBF, Polynomial, linear and sigmoid. The VGG-19 + SVM with polynomial kernel performs well for all classes (Benign, Malignant, and Normal) shown in figure 6 (D). The class 'normal' correctly classified instances are 95 only out of 130 images with near accuracy of 90%, and 154 malignant image features are classified correctly out of 210 (nearly 83% accuracy). Least performed class is benign with only 80% of accuracy. Figure 6 (A) presents the VGG-19+SVM with kernel RBF and the class 'normal' is classified highly with 87% of accuracy than other class accuracies. Figure 6 (B) shows the VGG-19+SVM with kernel Linear classifies the class 'normal' correctly 82 out of 130, 310 benign class images out of 440, and 105 images out of 210 in malignant. Figure 6 (C) shows the VGG-19+SVM with kernel sigmoid classifies the class 'normal' correctly 80 out of 130, 348 benign class images out of 440, and 163 images out of 210. Intrusting observation that the class malignant classifies 163/210 by the SVM sigmoid classifier than other classifiers, in other hand benign and normal class images are highly classified by the SVM-polynomial classifier.

SVM (RBF)					
Predicted					
Class	Benign	Malignant	Normal	Total	
Actual	Benign	370	39	31	440
	Malignant	60	144	6	210
	Normal	41	10	79	130
Total		471	193	116	780
(A) VGG-19+SVM(RBF)					
SVM (Linear)					
Predicted					
Class	Benign	Malignant	Normal	Total	
Actual	Benign	310	69	61	440
	Malignant	93	105	12	210
	Normal	41	7	82	130
Total		444	181	155	780
(B) VGG-19+SVM(Linear)					
SVM (Sigmoid)					
Predicted					
Class	Benign	Malignant	Normal	Total	
Actual	Benign	198	158	84	440
	Malignant	33	163	14	210
	Normal	23	27	80	130
Total		254	348	178	780
(C) VGG-19+SVM(Sigmoid)					
SVM (Polynomial)					
Predicted					
Class	Benign	Malignant	Normal	Total	
Actual	Benign	348	57	35	440
	Malignant	47	154	9	210
	Normal	27	8	95	130
Total		422	219	139	780
(D) VGG-19+SVM(Polynomial)					

Figure 8: Inception V3 + SVM Confusion matrix Analysis

Table 4: VGG-19+SVM (Kernel) Performance Parameters Analysis

VGG-19+SVM(Kernel)	Target Class	CA	F1	Precision	Recall	AUC
SVM (RBF)	Begin	0.7513	0.7770	0.786	0.7682	0.8592
	Malignant	0.8385	0.6986	0.7019	0.6952	0.8856
	Normal	0.8795	0.6544	0.6268	0.6846	0.9112
	All (Avg.)	0.7346	0.7355	0.7369	0.7346	0.8684
SVM (Polynomial)	Begin	0.8013	0.8229	0.8276	0.8182	0.8713
	Malignant	0.8538	0.7299	0.7264	0.7333	0.8960
	Normal	0.9013	0.7072	0.6992	0.7154	0.9215
	All (Avg.)	0.7782	0.7785	0.779	0.7782	0.8884

SVM (linear)	Begin	0.7295	0.7561	0.7694	0.7432	0.7768
	Malignant	0.7923	0.6087	0.6176	0.6000	0.8333
	Normal	0.8526	0.5907	0.5497	0.6385	0.8672
	All (Avg.)	0.6872	0.6888	0.6919	0.6872	0.8128
SVM (Sigmoid)	Begin	0.6269	0.6176	0.7321	0.5341	0.7094
	Malignant	0.7269	0.6019	0.4954	0.7667	0.8069
	Normal	0.8385	0.5227	0.5149	0.5308	0.8525
	All (Avg.)	0.5962	0.5976	0.6322	0.5962	0.7720

Table 4 shows the performance parameters, such as classification accuracy (CA), precision, F-1 values, recall, and AUC, of IV3+SVM (RBF, polynomial, linear, and sigmoid kernels) for the breast cancer classes normal, malignant, and benign. Intriguing findings from the analysis: 1. IV3 transfer learning developing features are very accurate for the cancer class "normal" (all SVM kernel models performed above 90% accuracy for the class "normal"); 2. In particular, SVM with the polynomial kernel classified all classes more highly than other kernels+SVM models, with SVM+sigmoid performing particularly poorly compared to the other models.

Figure 9 shows the confusion matrices of the VGG-19+SVM (kernels) 10-fold cross-validation models. As per the analysis, the extracted features from VGG-19 transfer learning were classified using an SVM classifier with various kernels like RBF, Polynomial, linear, and sigmoid. The VGG-19 + SVM with polynomial kernel performs well for all classes (Benign, Malignant, and Normal) shown in figure 6 (D). The class 'normal' correctly classified instances are 95 only out of 130 images with near accuracy of 90%, and 154 malignant image features are classified correctly out of 210 (nearly 83% accuracy). The least performed class is benign with only 80% of accuracy. Figure 8(A) presents the VGG-19 +SVM with kernel RBF and the class 'normal' is classified highly with 87% of accuracy than other class accuracies. Figure 8(B) shows the VGG-19 +SVM with kernel Linear class the class 'normal' correctly 82 out of 130, 310 benign class images out of 440, and 105 images out of 210 in malignant. Figure 8(C) shows the VGG-19 +SVM with kernel sigmoid classifies the class 'normal' correctly in 80 out of 130, 348 benign class images out of 440, and 163 images out of 210. The entrusting observation is that the class malignant classifies 163/210 by the SVM sigmoid classifier more than other classifiers, in another hand benign and normal class images are highly classified by the SVM-polynomial classifier.

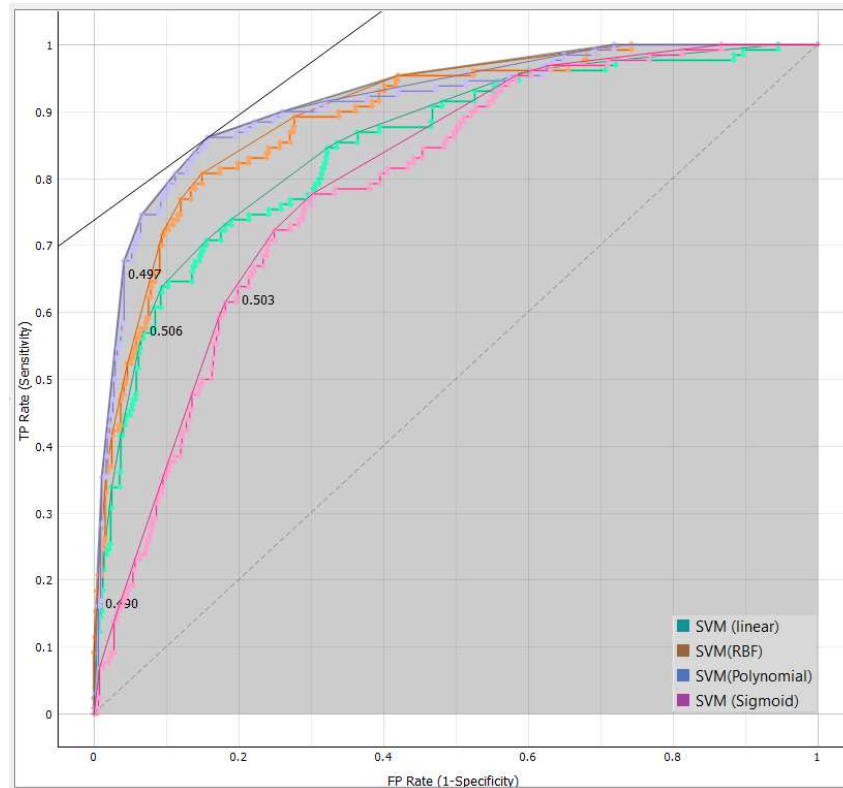


Figure 9: VGG-19 + SVM (Kernels) AUC under ROC Analysis

4.3.VGG-16 + SVM classification analysis:

Figure 10 shows the confusion matrices of the VGG-16+SVM (kernel) with 10-fold cross-validation models. In this analysis, the extracted features from VGG-16 transfer learning were classified using an SVM classifier with various kernels. The VGG-16 + SVM with a polynomial kernel performs better for all classes (Benign, Malignant, and Normal) shown in Figure 10 (D) than other kernels of SVM. The Normal class category images are classified as 93 out of 130 images with a near accuracy of 89%, and 154 out of 210, with a nearly 84% accuracy. Figure 10 (A) presents the VGG-16+SVM with a kernel RBF and the class name "Normal" correctly classified near 88%. Figure 10 (B) shows the VGG-16+SVM with a kernel linear classifying the class name "Normal," which correctly classified 83 out of 130, 327 images of the benign class out of 440, and 126 images out of 210. Figure 10 (C) shows the VGG-16+SVM with a kernel sigmoid classifying the class 'normal' correctly with 69 out of 130, 235 images of the benign class out of 440, and 154 images out of 210.

SVM (RBF)		Predicted			Total
Class		Benign	Malignant	Normal	
Actual	Benign	338	54	48	440
	Malignant	59	146	5	210
	Normal	33	8	89	130
Total		430	208	142	780

(A) VGG-16+SVM(RBF)

SVM (Linear)		Predicted			Total
Class		Benign	Malignant	Normal	
Actual	Benign	327	64	49	440
	Malignant	65	126	19	210
	Normal	33	14	83	130
Total		425	204	151	780

(B) VGG-16+SVM(Linear)

SVM (Sigmoid)		Predicted			Total
Class		Benign	Malignant	Normal	
Actual	Benign	235	149	56	440
	Malignant	40	161	9	210
	Normal	46	15	69	130
Total		321	325	134	780

(C) VGG-16+SVM(Sigmoid)

SVM (Polynomial)		Predicted			Total
Class		Benign	Malignant	Normal	
Actual	Benign	360	46	34	440
	Malignant	50	154	6	210
	Normal	25	12	93	130
Total		435	212	133	780

(D) VGG-16+SVM(Polynomial)

Figure 10: VGG-16 + SVM Confusion matrix Analysis

The figure 11 shows the AUC (area under the ROC curve) of the SVM (RBF), represented by the brown color, the SVM (Polynomial), indicated by blue, the SVM (Linear), chosen in green, and the SVM (Sigmoid), chosen in violet, with AUC values of 0.8625, 0.8747, 0.7639, and 0.7711, respectively. According to the AUC ROC analysis, all kernel SVM models perform well except for the sigmoid SVM model. The SVM (polynomial) performs best among the other SVM kernel models, with a value of 0.8747.

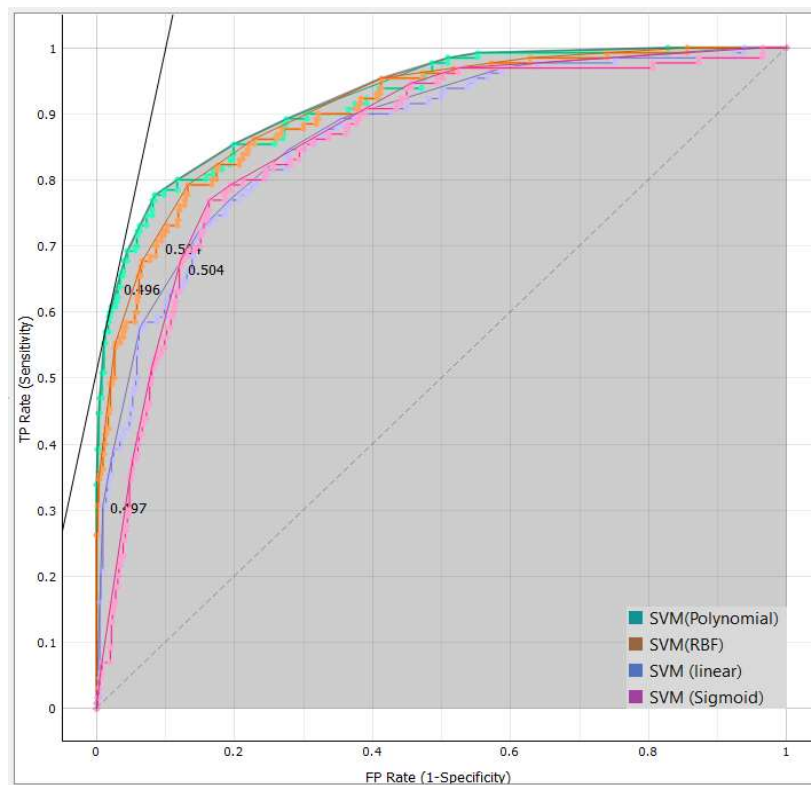


Figure 11: VGG-16 + SVM (Kernels) AUC under ROC Analysis

The table 5 shows the performance parameters such as classification accuracy (CA), precision, F-1 values, recall, (respective formulas shown in table 7) and AUC of VGG-16 + SVM (RBF, polynomial, linear, and sigmoid kernels) for the breast cancer classes Normal, Malignant, and Benign. Intriguing findings from the analysis include: 1. VGG-16 transfer learning developing features are very accurate for the cancer class "normal" (all SVM kernel models performed above 90% accuracy for the 'Normal' class); 2. In particular, SVM with the polynomial kernel classified all classes more highly than other kernels + SVM models. The SVM + sigmoid model performance is poor compared to the opposite kernels of SVM models.

Table 7: Performance metrics

Parameter	Formula
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$
F1	$2 * (Precision * Recall) / (Precision + Recall)$
Precision	$TP / (TP + FP)$
Recall	$TP / (TP + FN)$

Table 5: VGG-16+SVM (Kernel) Performance Parameters Analysis

VGG-16+SVM(Kernel)	Target Class	CA	F1	Precision	Recall	AUC
SVM (RBF)	Begin	0.7808	0.8123	0.7856	0.8409	0.8533
	Malignant	0.8526	0.7146	0.7461	0.6857	0.8746
	Normal	0.8872	0.6423	0.681	0.6077	0.8897
	All (Avg.)	0.7603	0.7577	0.7575	0.7603	0.8625
SVM (Polynomial)	Begin	0.7872	0.8074	0.8246	0.7909	0.8649
	Malignant	0.8449	0.7179	0.7032	0.7333	0.8723
	Normal	0.8987	0.7063	0.6835	0.7308	0.9071
	All (Avg.)	0.7654	0.7665	0.7684	0.7654	0.8747
SVM (linear)	Begin	0.6615	0.7014	0.6982	0.7045	0.708
	Malignant	0.7679	0.5371	0.5801	0.5000	0.7739
	Normal	0.8449	0.5754	0.5290	0.6308	0.8392
	All (Avg.)	0.6372	0.6361	0.6382	0.6372	0.7639
SVM (Sigmoid)	Begin	0.6179	0.5706	0.7795	0.4500	0.7390
	Malignant	0.7026	0.5842	0.4684	0.7762	0.8070
	Normal	0.8103	0.5195	0.4494	0.6154	0.7987
	All (Avg.)	0.5654	0.5658	0.6407	0.5654	0.7611

4.4.Comparative Analysis:

As compared to other experimental models, Inception V3 net + SVM (polynomial) with the 10-fold method performs well with a CA of 0.8744 and an AUC value of 0.9444. Table 4 shows the detailed analysis of all transfer learners' performances with SVM and all kernels' performances. The table 6 shows the comparative analysis of with all TLs + all kernel of SVM.

Table 6: TLs + SVM (Kernel) Comparative Analysis

+	Inception v3		VGG-19		VGG-16	
SVM (Kernel)	CA	AUC	CA	AUC	CA	AUC

SVM (linear)	0.8346	0.9252	0.6872	0.8128	0.6372	0.7639
SVM(Polynomial)	0.8744	0.9440	0.7782	0.8884	0.7654	0.8747
SVM(RBF)	0.8487	0.9359	0.7346	0.8684	0.7603	0.8625
SVM (Sigmoid)	0.7833	0.8991	0.5962	0.7720	0.5654	0.7611

5. Discussions:

In this part, we discuss our proposal model in comparison to other competitive models and existing works. After discussing and comparing transfer learning models (VGG-16, VGG-19, and Inception V3) with the SVM classifier, it was found that the Inception V3+SVM performed well with 0.8744 accuracy and 0.9444 AUC values with 10-fold cross-validation, better than other hybrid models. Figure 12 shows a detailed description of classification performance parameters, such as CA and AUC values, for each transfer learner (Inception-V3, VGG-19, and VGG-16), with SVM classifiers and breast cancer imaging.

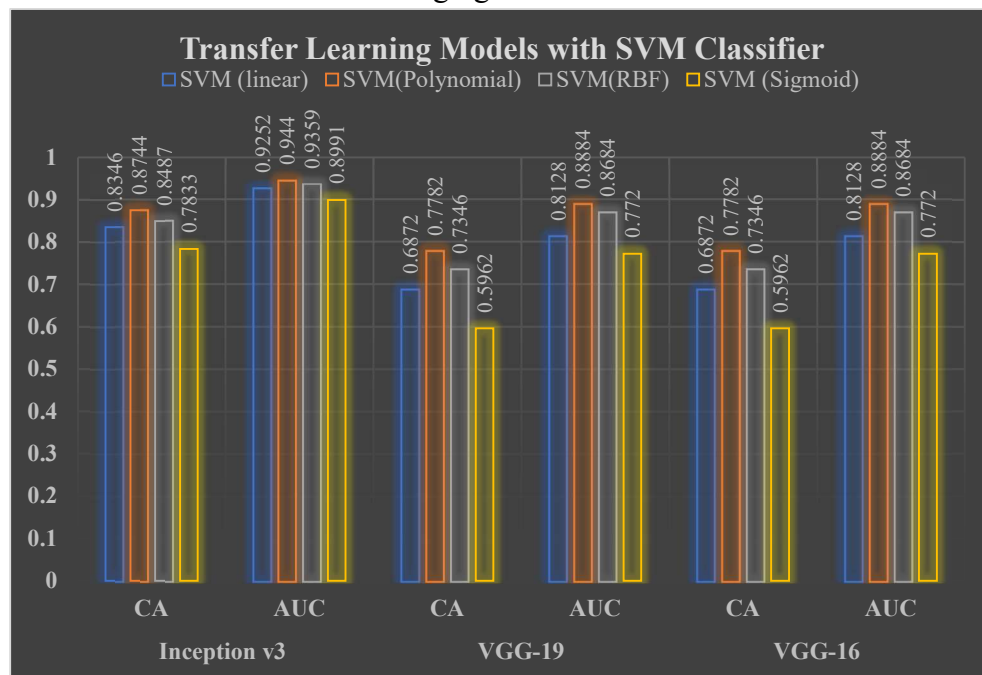


Figure 12: VGG-19 + SVM (Kernels) AUC under ROC Analysis

Vesal et al. (2018) [24] analyzed and classified breast cancer using transfer learning topologies Inception-V3 and ResNet50 networks on histology images(BACH 2018 Kagle data repository). Classification performance value evolved with 3-fold cross-validation. The testing average accuracy of Inception-V3 is higher than the ResNet50 network. Ferreira et al. (2018) [25] classified breast cancer using transfer learning (pre-trained) Inception-V3 network on histology images (BACH 2018 Kagle data repository). Classification performance value evolved with 3-fold cross-validation. The blind test set accuracy of Inception-V3 is 76%.

We identified some incorrectly classified images in all hybrid models: 11, 55, 67, 74, 121, etc. in the malignant category; 30, 62, 76, 258, 372, 425, etc. in the benign class; and 17, 113, 114, 116, etc. in the "normal" category of breast cancer. Figure 13 shows some of these unclassified images.

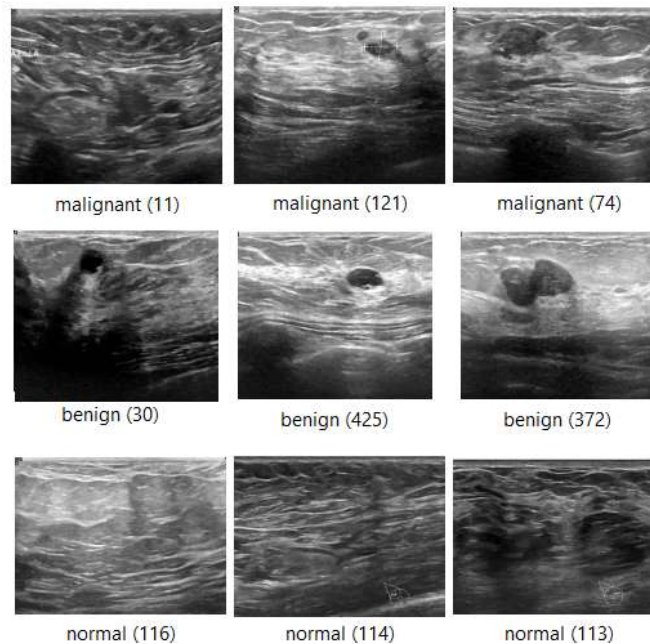


Figure 13: Samples of unclassified Breast cancer Ultrasound images

6. Conclusion

Breast cancer is the most critical and suspicious disease mostly in women. Early detection of disease is essential for the patient's protection and survival. Transfer Deep learning models are useful for extracting the features of breast cancer ultrasound images. The system employs deep transfer learners that are pre-trained on large datasets to learn the features of breast cancer images. The deep transfer learners then transfer their learned features to a SVM classifier model, which is trained on a smaller dataset of breast cancer images to classify them as benign, malignant or normal. SVM classifier with all kernel models are known for their ability to handle high-dimensional data and classify data accurately, making them suitable for breast cancer identification. In this research, we used Inception-v3, VGG-16 net, and VGG-19 net for feature extraction of ultrasound breast cancer images and classify with SVM (polynomial, sigmoid, RBF and linear). The SVM polynomial kernel performs better for all features of transfer learning than the remaining SVM kernels. According to our findings, the Inception-v3+SVM (polynomial) performance is very high, with a 0.944 AUC and 87.44% classification accuracy, higher than other experimental models. The proposed system has the potential to improve breast cancer identification accuracy, which is crucial for early detection and treatment of the disease.

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